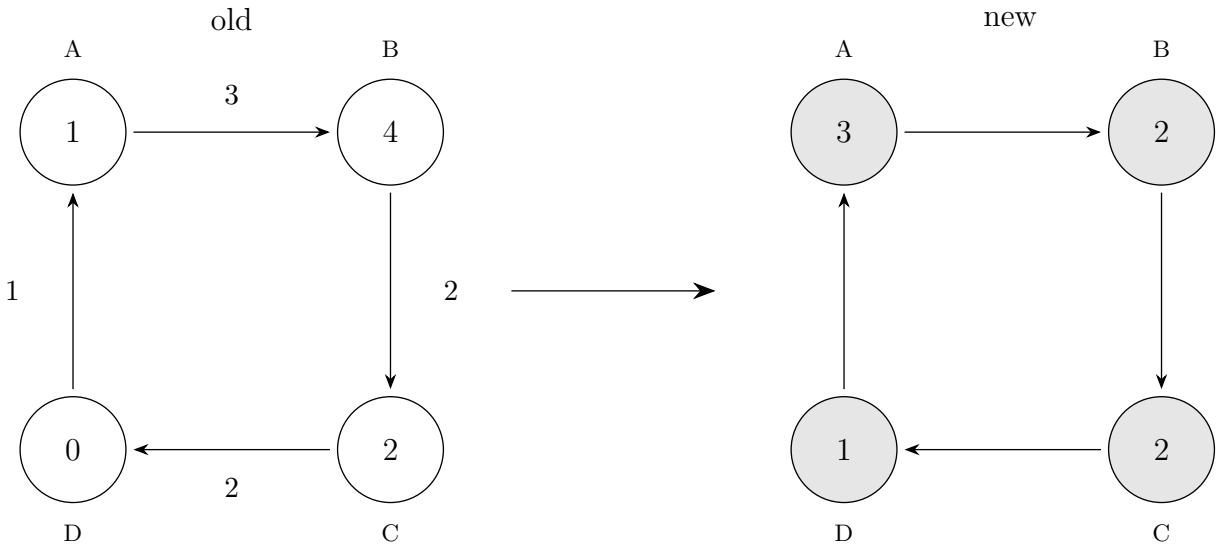
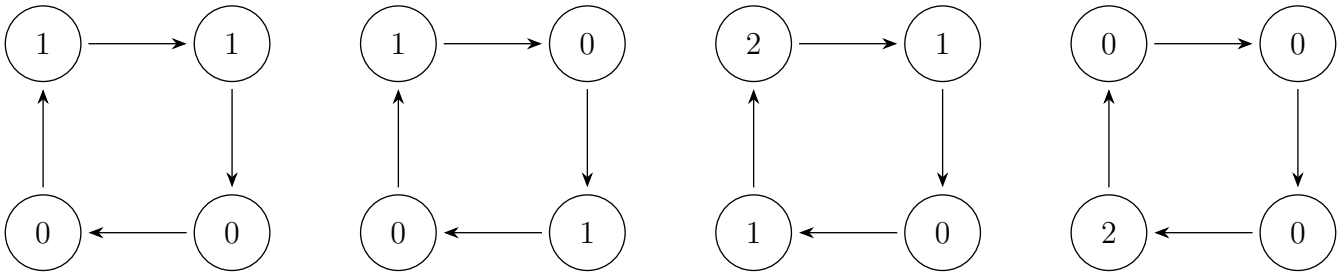
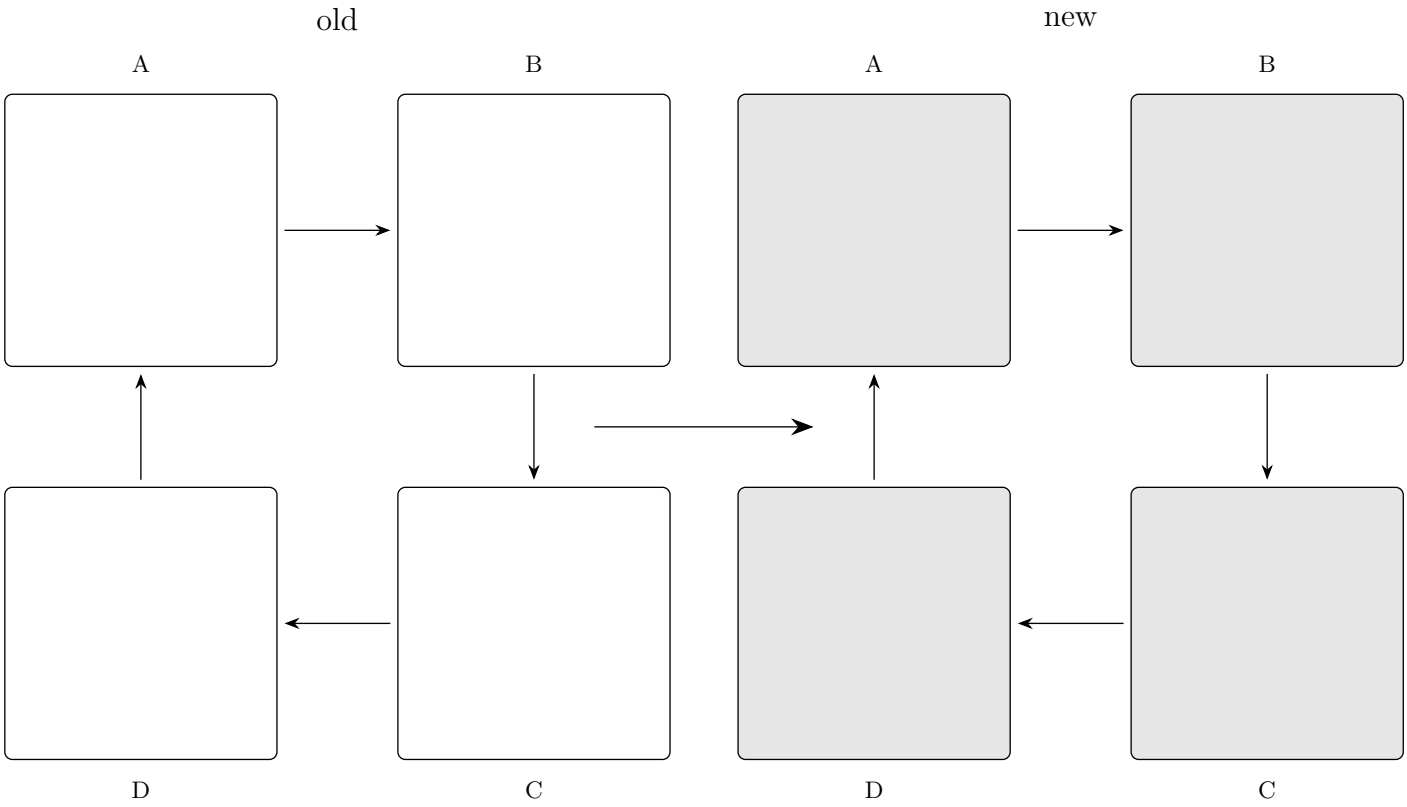
Use whole-number heights, including 0. Follow each arrow: the gap between its two old heights becomes the new height at its tail. Make every new tower before changing any old tower. An all-zero start takes 0 moves.



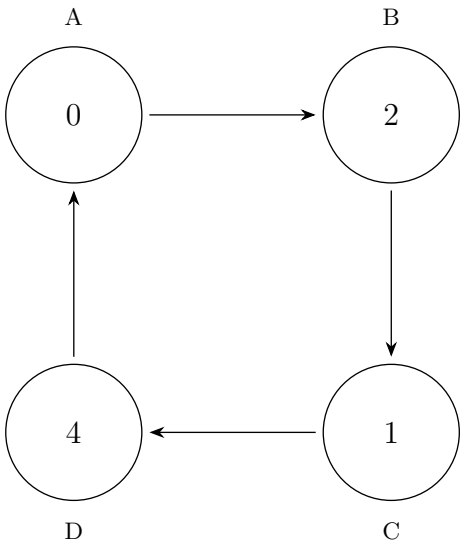
Edge A to B has gap 3, so new A is 3.

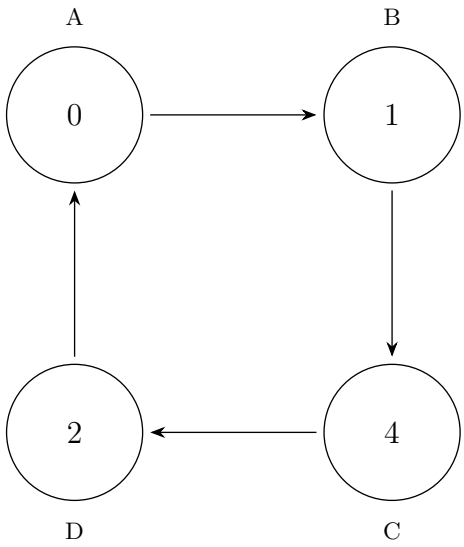
Problem 1: Run each start until all four heights are 0. Which lasts longest?



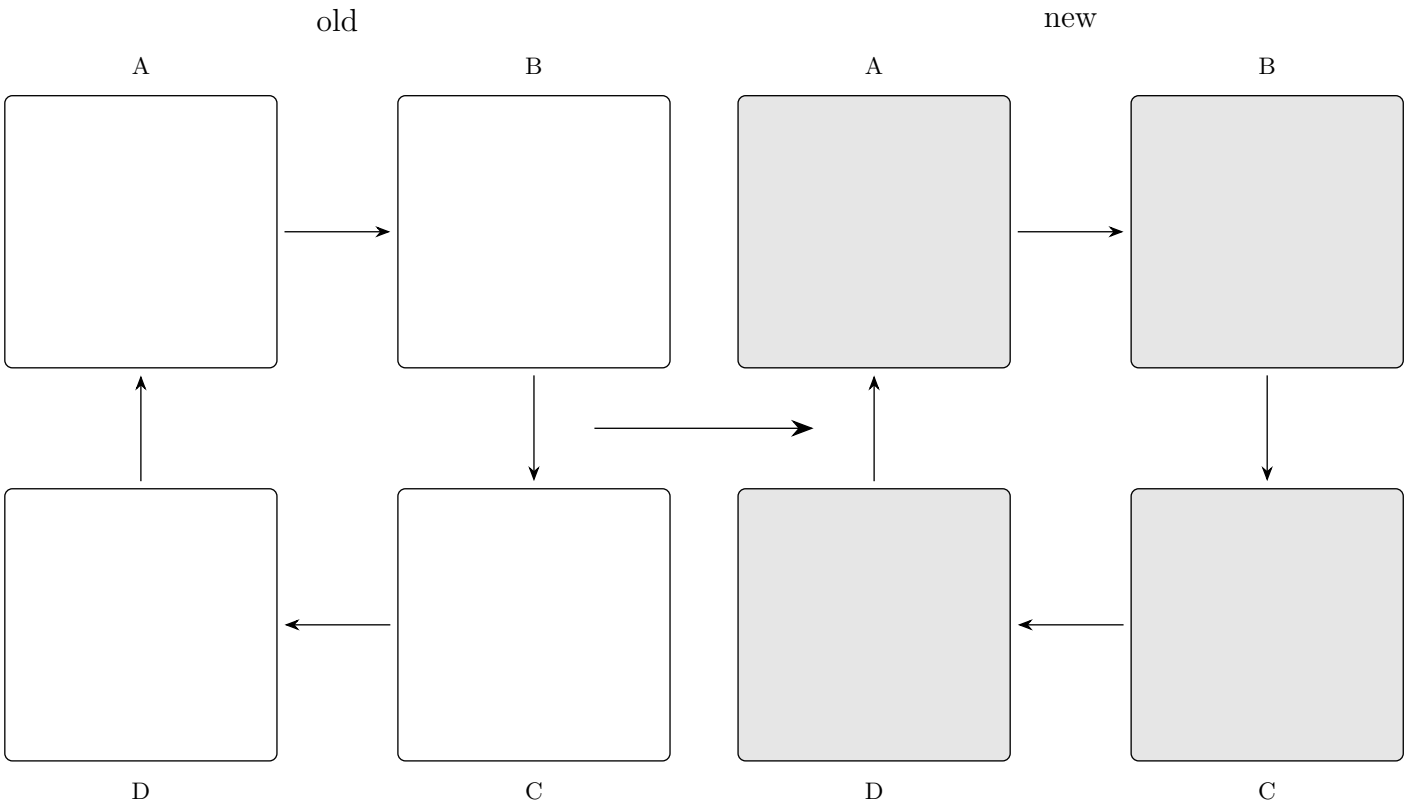


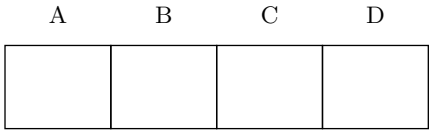
Problem 2: Try different orders of 0, 1, 2, and 4. Which order lasts longest?



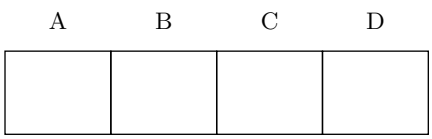


Problem 3: Choose starting heights from 0 to 5. Find a ring that lasts as many moves as you can. Count the move that first makes all four heights 0.



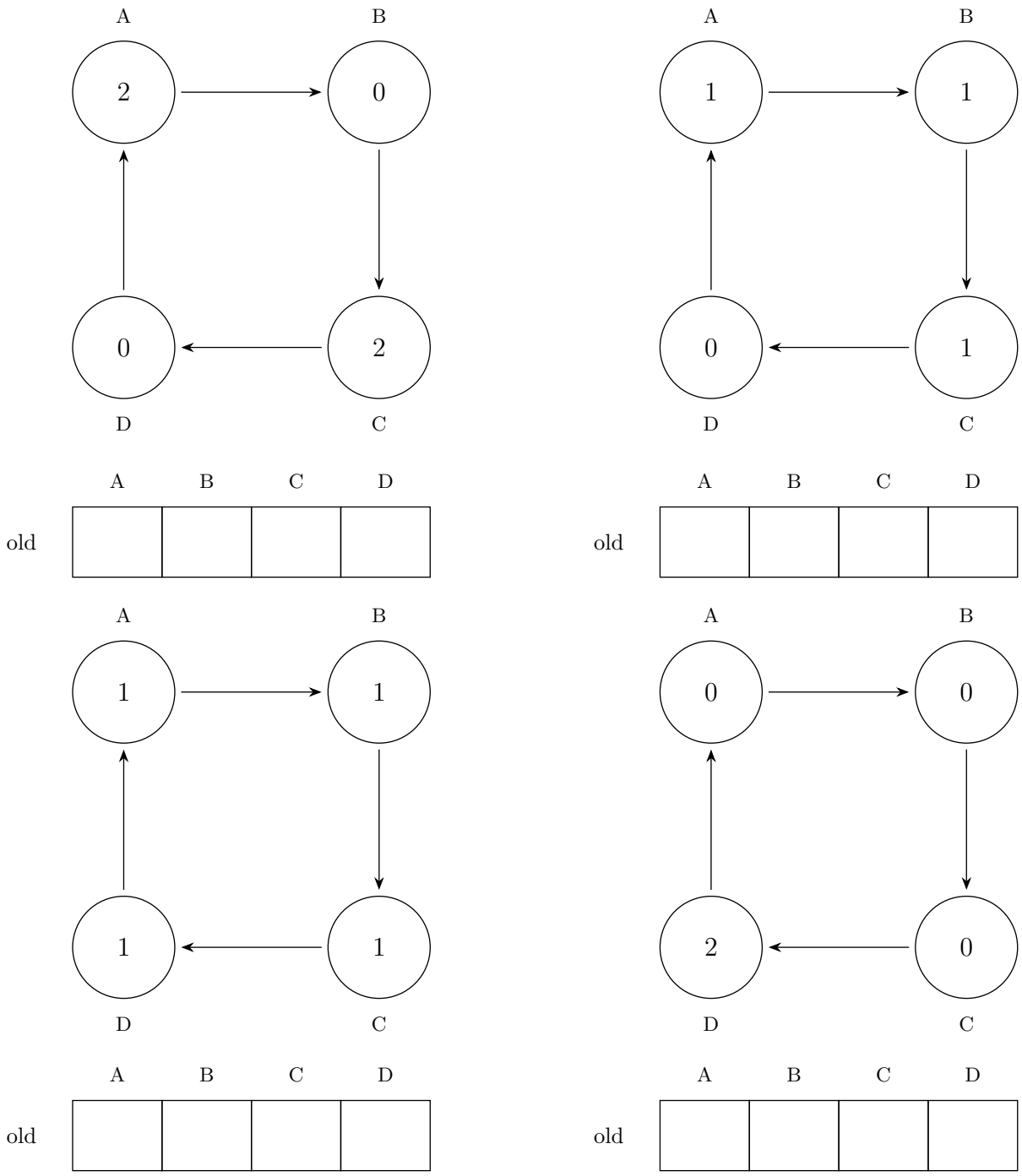


Moves to zero: _____

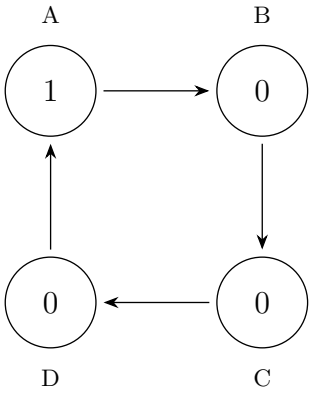


Moves to zero: _____

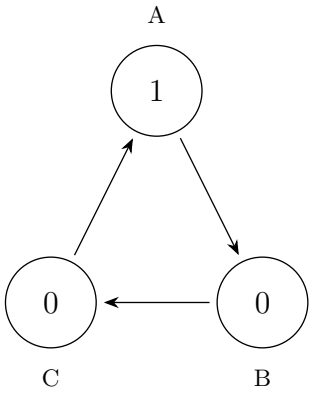
Problem 4: Could each ring be the result of one move? Find an old ring that makes it, or explain why no old ring can work.



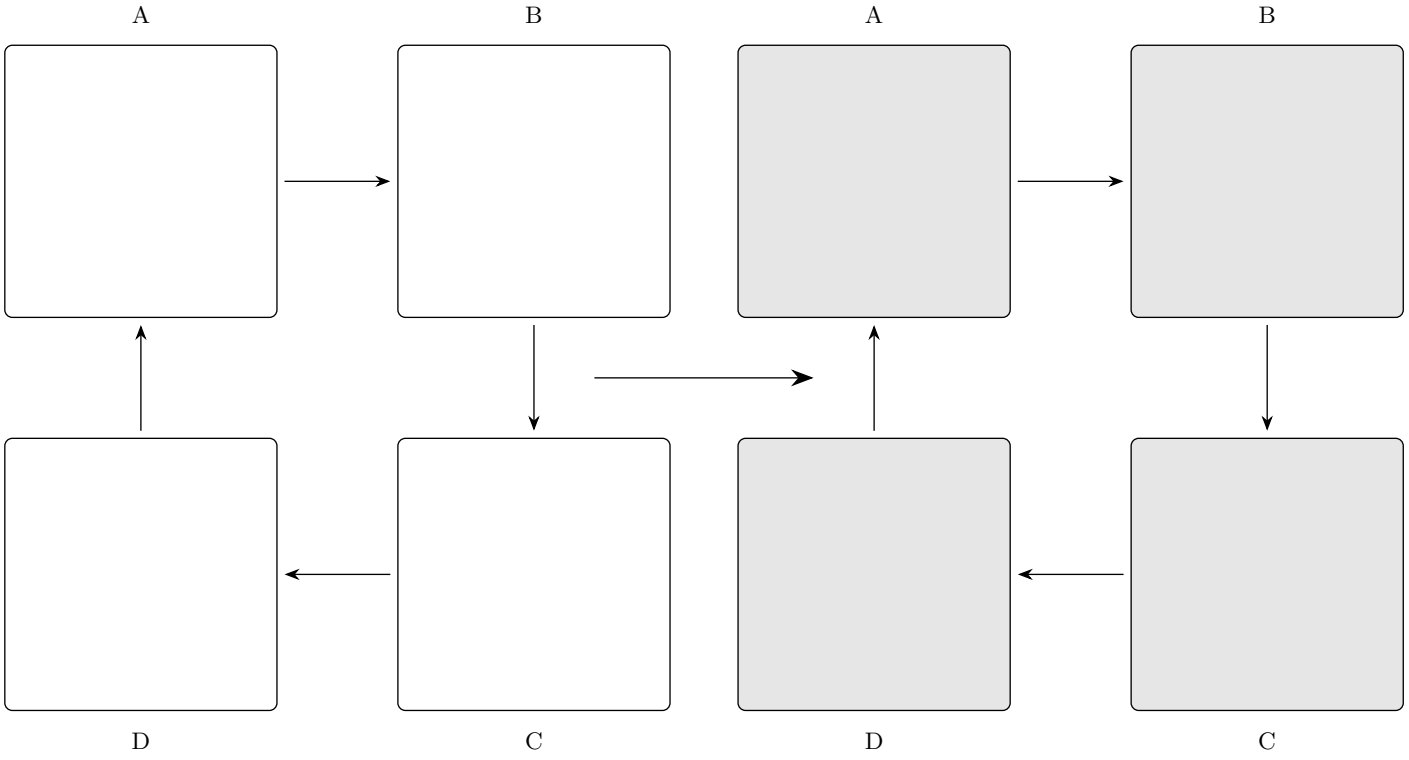
Problem 5: Use the gap rule on both rings. Keep going until the whole ring repeats or becomes all zeros. Does the number of stations change what can happen?



old

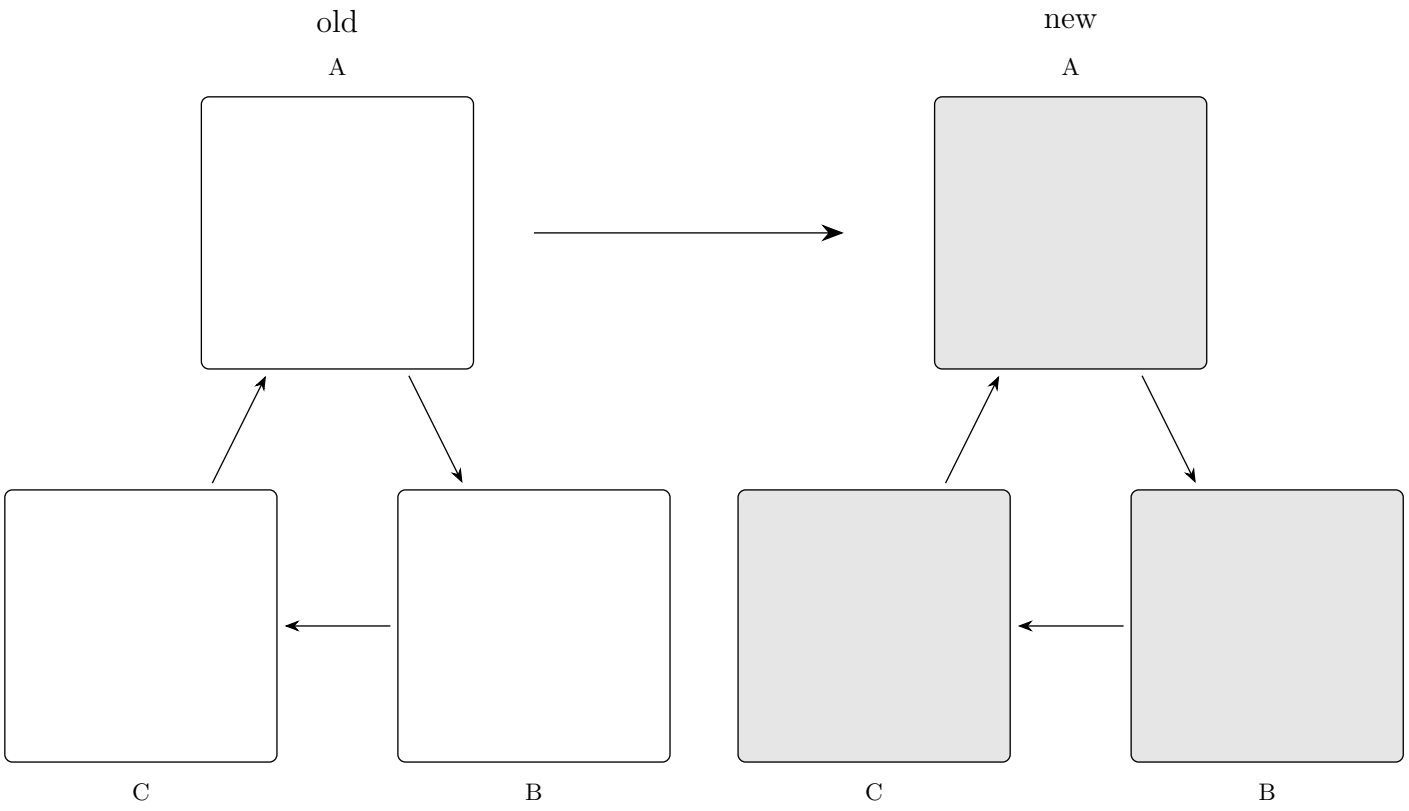


new



	A	B	C	D
1				
2				
3				

	A	B	C	D
4				
5				
6				



	A	B	C
1			
2			
3			

	A	B	C
4			
5			
6			

Problem 6: Can a new tower be taller than every old tower? Can the tallest height stay the same for a move? Give an explanation or an example for each question.

